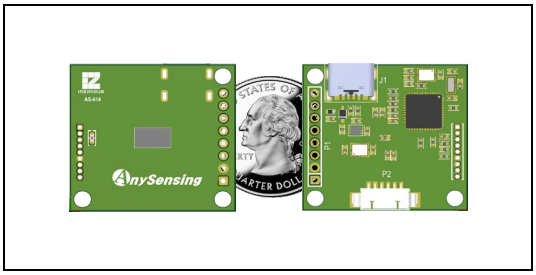
AS_614 with AI

Distance, Location, Human Vital-sign detection multi-channel FMCW Radar
with High performance, Small size, Low-power
GPIO, UART, Indicate LED

User Manual

1 Features

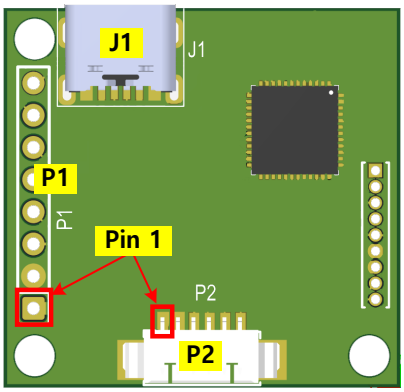


- 60GHz multi-channel FMCW radar
- One-board solution
- Include antenna and DSP
- Wide input voltage available: 3.3 to 7.0V
- Compact size and easy to use
- GPIO and UART interface
- Low power consumption: 700 mW
- Provides a variety of sensing modes
  - Wall (side) mounting position measurement mode: maximum distance of 8m
  - Ceiling mounting position measurement mode: maximum diameter of 5m
  - Wall (side) mounting occupancy detection mode: maximum distance of 8m
  - Ceiling mounting occupancy detection mode: maximum diameter of 5m
  - Wall (side) mounting vital signal measurement mode: maximum distance of 2.5m
  - Wall (side) mounting combined mode for vital signal and position measurement: Vital signal up to 3m, position measurement up to 8m
  - Raw Data Output Mode: Outputs unprocessed raw data on a per-antenna-channel basis (optional)
  - Advanced vital sign mode: Includes HRV data, such as RMSSD and SDNN (option)
  - Restroom ceiling mounting mode (option)
  - Hospital ceiling mounting patient monitoring mode (option)
- Three types of power source inputs available: USB-C, 2.54mm pin header, and connector

2 Specification

Content		Definition
Model name		AS_614
Hardware size		32.0 x 30.0 x 3.35 mm (No connector) 32.0 x 30.0 x 5.1 mm (Applying connector)
Weight		3.07 g
Transmit frequency		60GHz
Antenna		1 TX (EIRP : 10dBm), 4 RX / Antenna gain : 3dBi
RF pattern (-3dB width)		Horizontal 100° / Vertical 100°
Supply voltage / current		3.3 ~ 7 VDC / max 700 mW
Operating / Storage temperature		-40 ~ 85 ℃ / -55 ~ 125 ℃
Interface		1 GPIO, 1 UART(TTL level),
Function		Motion, Distance, Vital-sign(BR, HR, HRV), Indicate LED
Distance	Range	Distance : max. 8m Vital sign : max. 3m
	Resolution	Distance : 12 cm Vital sign : 85%

3 Pin description



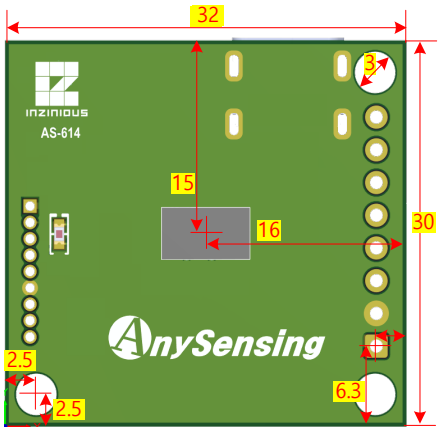
AS_614

PIN	Name	P1	P2
1	VCC	Power + (3.3 ~ 7V)	Power + (3.3 ~ 7V)
2	GND	Power -	Power -
3	GPIO	out / in	out / in
4	NRST	N.C or High : Normal operation Low : Reset	N.C or High : Normal operation Low : Reset
5	UART TX 1	UART 1 data transmit	UART data transmit
6	UART RX 1	UART 1 data receive	UART data receive
7	N.C	Not used	Not used
8	N.C	Not used	Not used

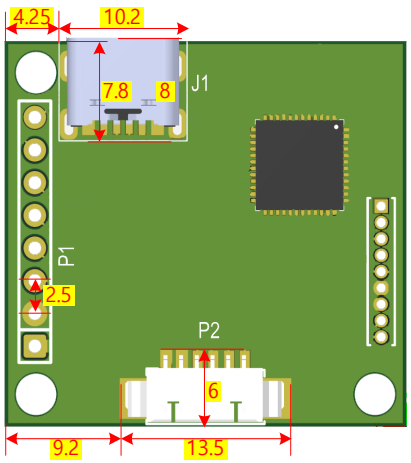
P1 : 2.54mm Pin header
P2 : 12505WR-06, The connecting connector is 12505HS-06.
J1 : The USB-C type connector is used solely for power supply.

- The GPIO port is set to output by default.
- If GPIO is set to output, the state in which the object is detected is output through GPIO.
- If GPIO is set to input, the last character in the data output is used for output.

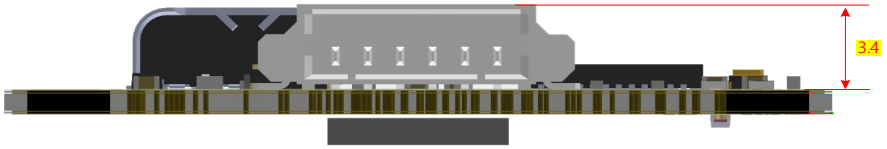
4 Dimension



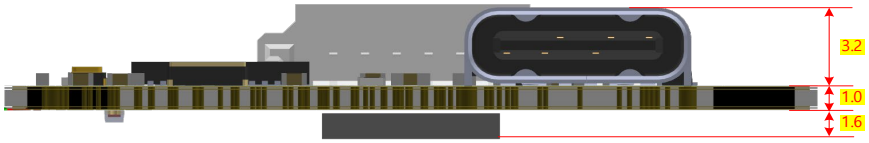
Top



Bottom



Install connector (12505WR-06)



Install USB-C connector

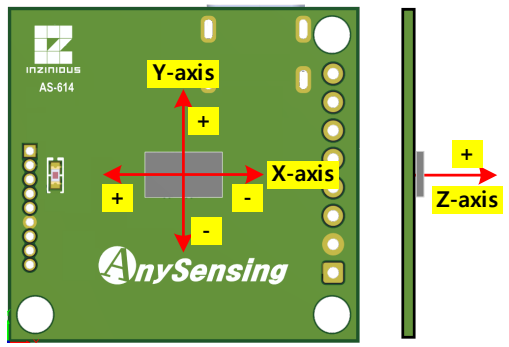
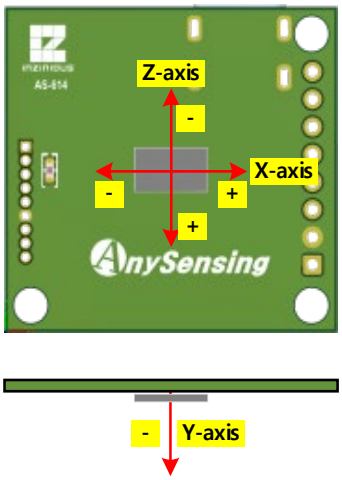
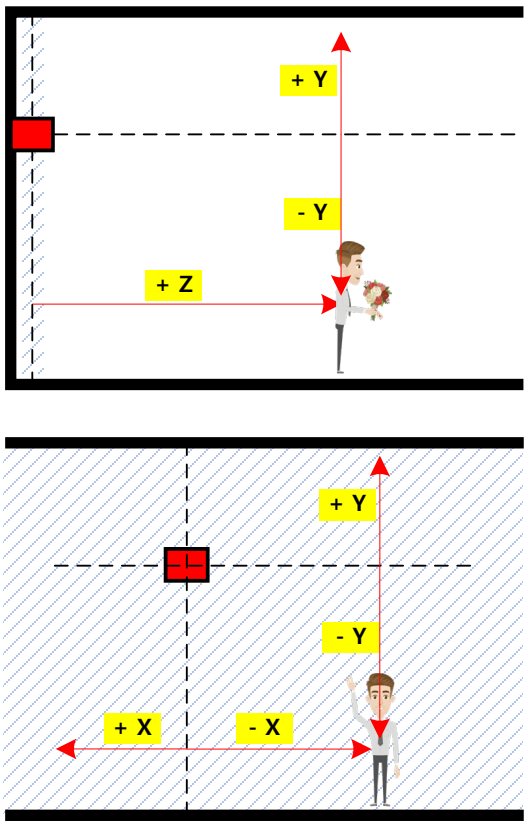
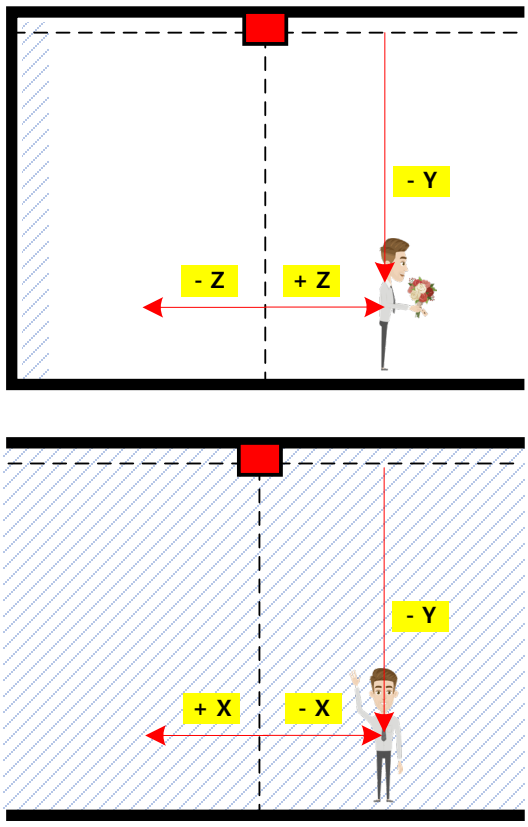
5 Function

Default Mode	Function
Wall mounting position measurement mode	**Mode 1:** Default setting mode: Installed on the wall, it detects the positions of people or moving objects. It provides data on the location of the detected object as well as information such as energy levels. It supports a normal mode that outputs only the X and Z coordinates and energy values of objects, as well as an extended mode that outputs the X, Y, and Z coordinates along with distance information to the object and energy values.
Ceiling mounting position measurement mode	**Mode 2:** Installed on the ceiling to detect people or moving objects on the floor. It outputs the location of the detected object as X and Z coordinates, or provides data for the detection area. It supports a normal mode that outputs only the X and Z coordinates and energy values of objects, as well as an extended mode that outputs the X, Y, and Z coordinates along with distance information to the object and energy values.
Wall mounting occupancy detection mode	**Mode 3:** Installed on the wall, it outputs the approximate number of people or moving objects. Supports a mode that outputs the approximate number of people in the entire space and a mode that divides the space into nine zones and outputs the approximate number of people in each zone.
Celling mounting occupancy detection mode	**Mode 4:** Installed on the ceiling, it outputs the approximate number of people or moving objects. Supports a mode that outputs the approximate number of people in the entire space and a mode that divides the space into nine zones and outputs the approximate number of people in each zone.
Wall or side mounting vital sign measurement mode	**Mode 5:** It detects the vital sign of an individual within 2.5 meters and provides outputs such as the individual's location, heart rate, respiratory rate, heart pulse, breathing waveform, and movement data. The system supports a normal mode that outputs only vital sign and distance information, and an extended mode that also includes location details.
Wall mounting vital sign and position measurement combined mode	**Mode 6:** A complex mode that detects biological signals and location information. It detects location, heart rate, and respiration up to 3m, detects location and respiration up to 5m, and only detects location up to 8m.
Option Mode	
Raw Data Output Mode	**Mode 10:** Provides Real and Imaginary data for each channel of the MMIC.
Advanced vital sign detection mode	**Mode 11:** The Advanced Vital Sign Detection Mode provides additional HRV data, including RMSSD and SDNN, in the extended mode of Mode 5.
Restroom ceiling mounting mode	**Mode 12:** Installed on the bathroom ceiling, it outputs the occupancy status and breathing information of individuals in the bathroom. It provides the ability to distinguish whether a person is standing, sitting, or lying down.
Ceiling mounting hospital bed mode	**Mode 13:** Installed on the hospital ceiling, it provides functions to distinguish the patient's presence on the bed, including posture, heart rate, respiration, and HRV information, as well as whether the patient is standing, sitting, or lying down next to the bed. It also offers information on fall risks and pressure ulcers.

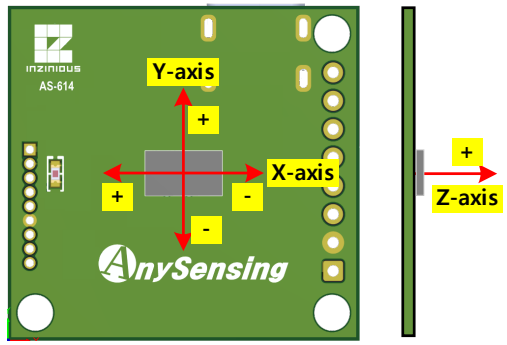
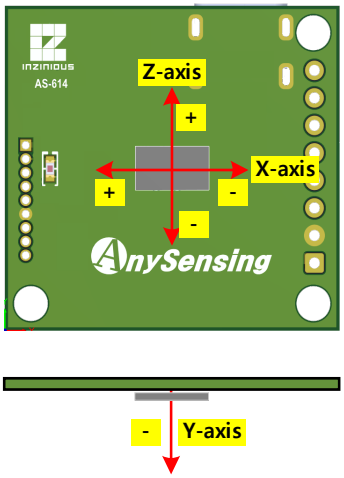
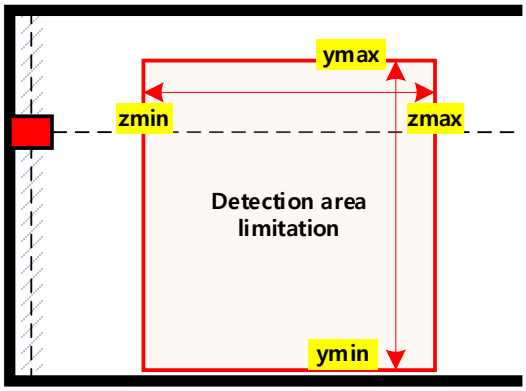
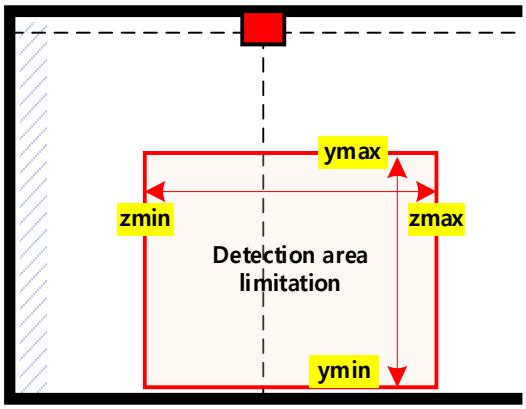
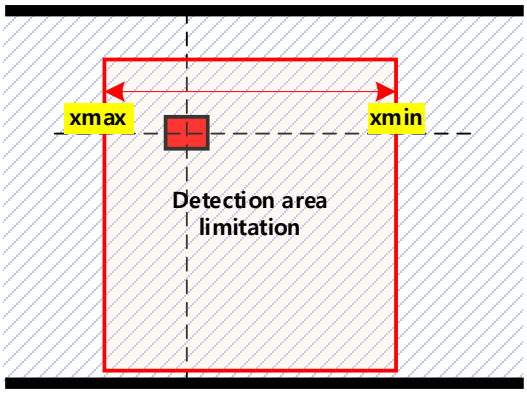
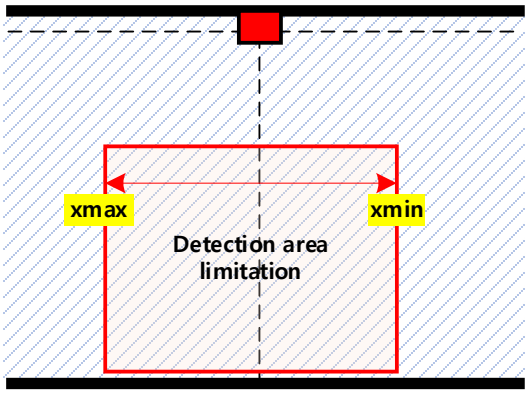
- If you need height information (Y-axis data), you can calculate it using X-axis data, Z-axis data, and linear distance data. The extended mode includes additional data such as Y and linear distance.
- After changing the mode, the first output data may not be normal because it is based on the data prior to the mode change.
- The GPIO output is set to a high state when the target is detected and resets to a low state when the target is not detected.

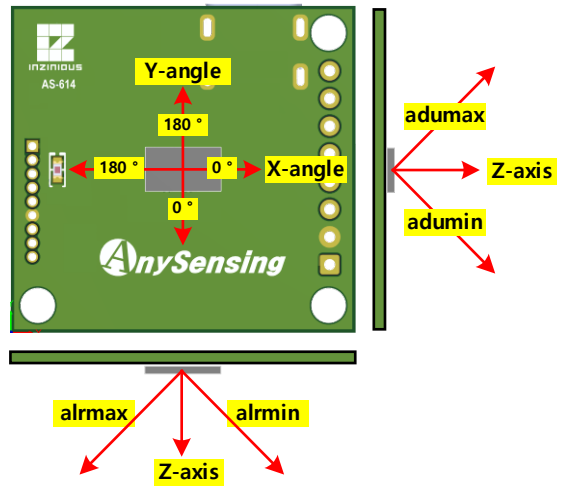
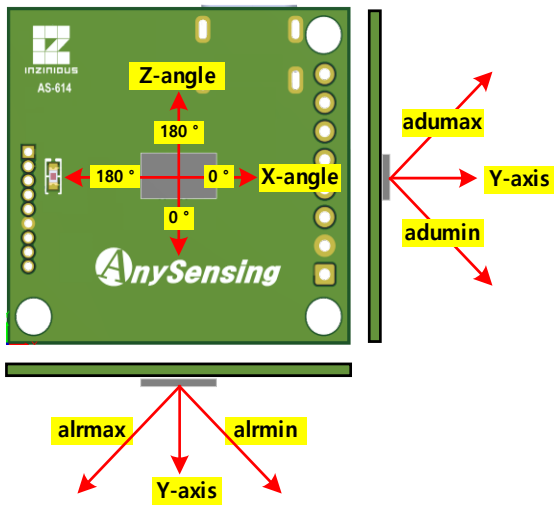
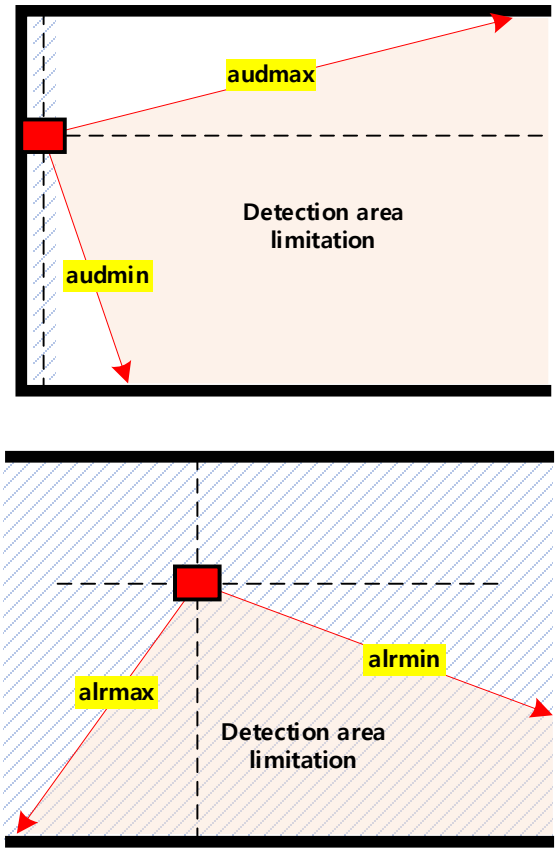
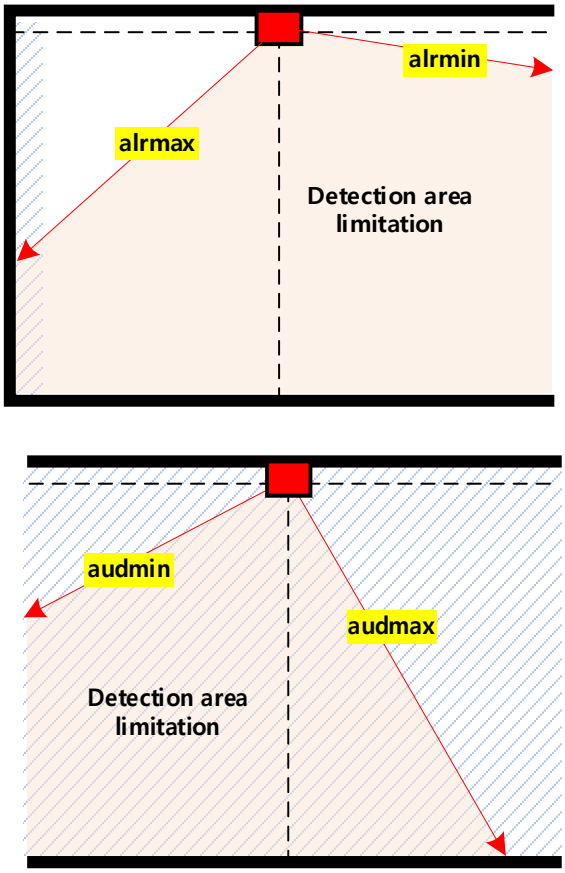
- The default mode includes all features of AS_614, while the option mode is activated through a separate licensing agreement.
- **In vital sign detection mode, the radar cannot distinguish and detect only a specific person. If a moving object is closer to the radar than the detected person within the detection area, it becomes difficult to differentiate between their movements, and the quality of the detected biometric signals cannot be guaranteed.**
- **The radar responds to various types of movements, including clock hands, air conditioners, fans, air purifiers, pets, and other similar objects. In particular, if a person being detected moves their arms or hands in front of their chest—such as using a mouse or keyboard, operating a smartphone, or crossing their arms—the biometric signals may not be detected properly.**

6 Radar coordinate orientation

<table>
<tr><th>Wall (side) installation</th><th>Ceiling installation</th></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td>X-axis : (-) Left, (+) Right
Y-axis : (-) Down, (+) Up
Z-axis : (+) Front</td><td>X-axis : (-) Left, (+) Right
Y-axis : (-) Hight
Z-axis : (-) Back, (+) Front</td></tr>
<tr><td colspan="2">The coordinates of AS_614 are output based on the presence of people, depending on the installation environment.</td></tr>
</table>

The detection performance in the X-axis direction is relatively better than that in the Y or Z-axis directions.

Wall (side) installation detection distance limit	Ceiling installation detection distance limit
	
	
	
Z-axis is always +	Y-axis is always -

Wall (side) installation detection angle limit	Ceiling installation detection angle limit
	
	

- The angle is set based on the radar, with a minimum range of 0 degrees and a maximum range of 180 degrees, and the sign is always positive (+).
- In the function that specifies the detection area, the angle settings are not clearly restricted compared to the distance settings. The angle range settings provide a general limitation of the range.

7 Mode Description

7.1 Wall mounting position measurement mode (mode 1)

- Installed on the side or wall next to the person or object you want to detect.

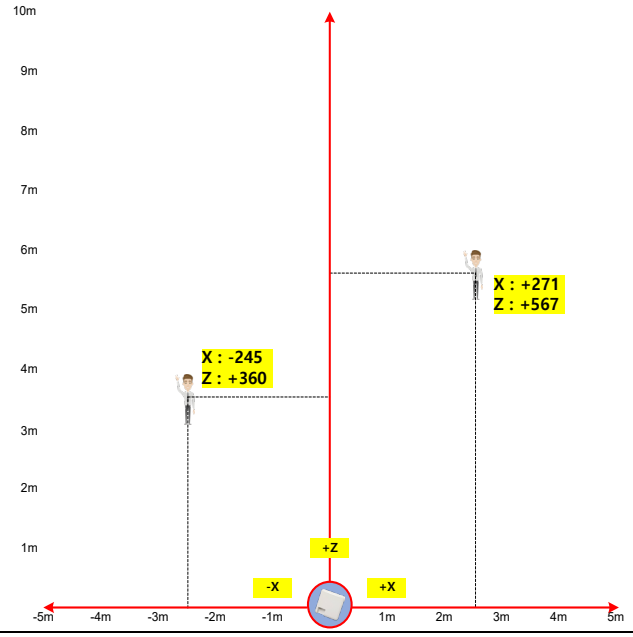
- The basic detection distance settings are as follows:

x_min	-4,000 mm	x_max	4,000 mm
y_min	-2,000 mm	y_max	1,500 mm
z_min	400 mm	z_max	8,000 mm
alrmin	0 °	alrmax	180 °
adumin	20 °	adwmzx	160 °
Z always has + values			

- The basic detection angle is 0 to 180° on the x-axis and 20 to 160° on the y-axis. The smaller the object, the narrower the detection angle.
-
- The default number of detected objects is 3, with a maximum of 5.
- The output method includes a mode that provides coordinates and size of movement, as well as a mode that outputs the detection area divided into zones.
- It offers a mode that outputs a maximum of 12 times per second and a mode that outputs once per second.

- **Detection is not possible or accuracy is low in the following cases:**
  - Depending on the installation environment and the type of object being detected, the detectable distance may be shorter than the set detection distance.
  - Stationary people cannot be detected; detection requires minimal movement. The minimum amount of movement includes breathing within 2m, conversation within 5m, typing on a keyboard within 7m, and walking slowly or shaking your shoulders beyond 7m.
  - Close distances within 3m can be detected even near 0° or 180°. As the distance increases, detection performance decreases, requiring greater movement to be detected.
  - Normal detection is not guaranteed in environments where furniture or fixtures are arranged in a complex manner. Detection may not be successful, or the results may be inaccurate. This is especially true in spaces with many metallic objects, where virtual images may be detected due to reflected waves. The optimal sensing environment is an open space, such as a conference room.
  - It detects all moving objects within the detection space. Specifically, it can detect running fans in devices such as air conditioners, electric fans, and air purifiers, as well as moving curtains or long dangling cables. It also detects moving pets. However, it cannot distinguish between people, animals, and objects, nor can it measure their sizes.
  - In the case of detecting a person, the measured result may continuously fluctuate within a range of 20 to 30 cm, even if the person is still.
  - Objects directly behind the object in front cannot be detected.
  - If a person is close by, they are detected as a single object. The degree of closeness varies depending on the distance from the radar; the farther away the person is, the more they must be separated to be recognized as two distinct objects.
  - Since the maximum detection distance varies depending on the environment and angle of the space where the radar is installed, the maximum detection distance and range must be verified and tested in a real environment.
  - The reliable detection range is approximately 5m for people and 10m for vehicles. For detecting distances beyond 10m, please contact us separately.

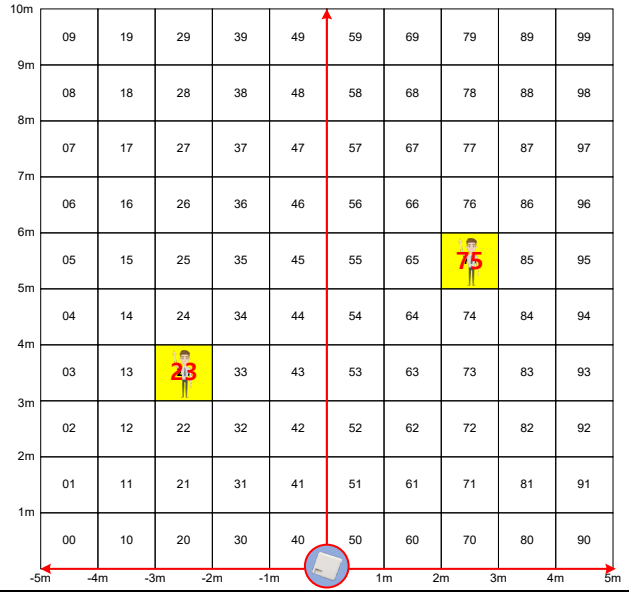
- Data Output Format: ASCII
  - Wall installation distance output mode:



NORMAL MODE		
#ABBBBCCCDDD! (13 byte) (default mode)		
Item	**Byte**	**Content**
#	1	Wall installation distance output mode data start character
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
BBBB	4	X coordinate displays the left and right positions based on the radar, with the unit in cm. The first byte indicates the sign: '-' for left and '+' for right.
CCC	3	Z coordinate, indicates the position in front and behind the radar, unit in cm
DDD	3	Amount of power in an object, Mark all values larger than 999 as 999
!	1	Character indicating the end of data.
#0!		When no object is detected, output at set time intervals
EXTENSION MODE		
#ABBBBCCCCDDDEEE! (17 byte) (default mode)		
Item	**Byte**	**Content**
#	1	Wall installation distance output mode data start character
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
BBBB	4	X coordinate displays the left and right positions based on the radar, with the unit in cm. The first byte indicates the sign: '-' for left and '+' for right.
CCCC	4	Y coordinate displays the downward and upward positions based on the radar, with the unit in cm. The first byte indicates the sign: '-' for downward and '+' for upward.
DDD	3	Z coordinate, indicates the position in front and behind the radar, unit in cm
EEE	3	Amount of power in an object, Mark all values larger than 999 as 999
!	1	Character indicating the end of data.
#0!		When no object is detected, output at set time intervals

- It does not display the sign because it only outputs the Z direction's positive distance.

■ Detection zone output mode



*ABC! (5 byte) (default mode)

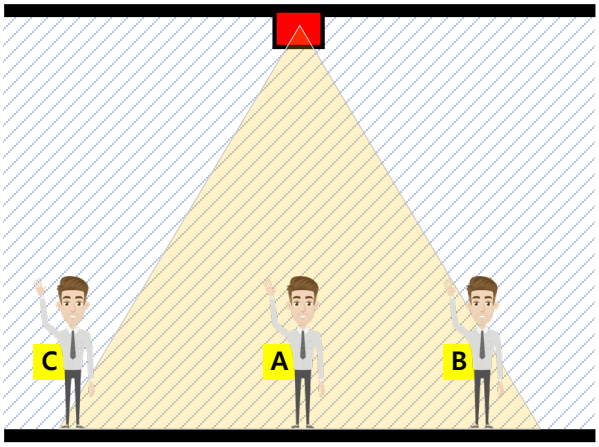
Item	Byte	Content
*	1	Starting character indicating the detection zone output mode
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
B	1	X-coordinate distance 0: Radar reference left: -400 to cm 1: Radar reference left: -300 to -399 cm 2: Radar reference left: -200 to -299 cm 3: Radar reference left: -100 to -199 cm 4: Radar reference left: -0 to -99 cm 5: Radar reference right: 1 to 99 cm 6: Right side based on radar: 100 to 199 cm 7: Radar reference right: 200-299 cm 8: Right side based on radar: 300-399 cm 9: Right side of radar: 400 to cm
C	1	Z-coordinate, the distance ahead 0: Radar reference forward: 0 to 99 cm 1: Radar reference forward: 100 to 199 cm 2: Radar reference front: 200-299 cm 3: Radar reference front: 300-399 cm 4: Radar reference forward: 400 to 499 cm 5: Radar reference forward: 500 to 599 cm 6: Radar reference front: 600 to 699 cm 7: Radar reference front: 700 to 799 cm 8: Radar reference front: 800 to 899 cm 9: Radar reference front: 900 to cm
!	1	Character indicating the end of data.
*0!		When no object is detected, output at set time intervals

7.2 Ceiling mounting position measurement mode (mode 2)

- Installed on the ceiling to detect the locations of people or objects on the floor.
- The basic detection distance settings are as follows:

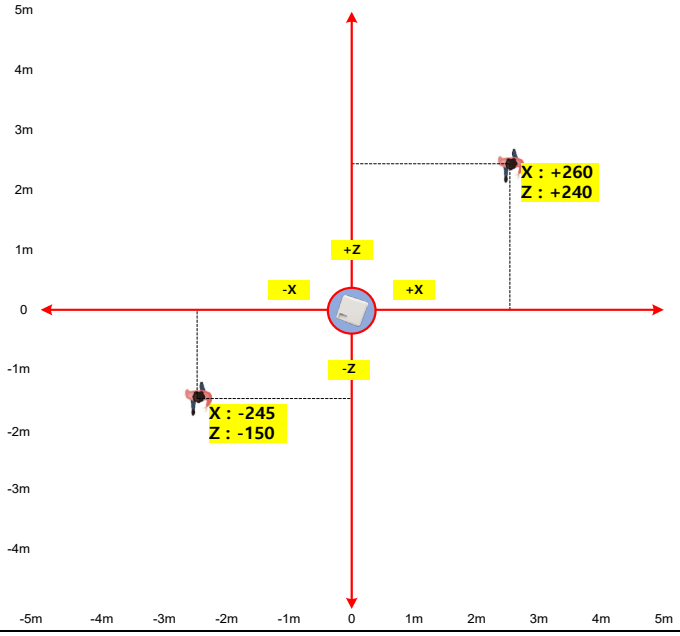
x_min	-3,000 mm	x_max	3,000 mm
y_min	-3,000 mm	y_max	-400 mm
z_min	-3,000 mm	z_max	3,000 mm
alrmin	15 °	alrmax	165 °
adumin	15 °	adwmzx	165 °
Y always has - values			

- The smaller the object, the detectable angle becomes narrower.
- Compared to the wall installation method, the detection range is shorter. The higher the ceiling where the radar is installed, the wider the detection range. Based on a 2.5m ceiling, it detects people within a detection range of about 4~5m in diameter.
- **Detection is not possible or accuracy is low in the following cases:**



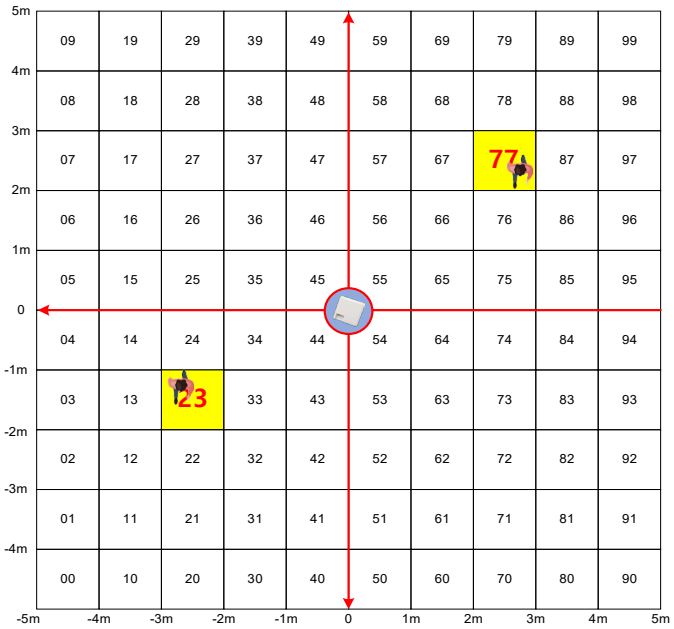
- Depending on the installation environment and the type of object being detected, the detectable distance may be shorter than the set detection distance.
- In the figure above, detection is possible in cases A or B, but not in case C, so the detectable range is up to B.
- Although the distance of A is closer to the radar than B, the reflection area is smaller than that of B, so small breathing or movements may not be detected.
- Since the maximum detection range depends on the height of the installation and the surrounding environment, it is necessary to conduct tests in the installation environment to determine the maximum detection distance.
- When multiple people are close together or side by side in the radar's direction, they are recognized as a single object.
- It does not function normally in environments where furniture is arranged complexly within the sensing area, especially in places with many metallic objects.

- Data Output Format : ASCII
  - ■ Ceiling installation distance output mode



NORMAL MODE		
$ABBBBCCCCDDD! (14 byte) (default mode)		
Item	**Byte**	**Content**
$	1	Wall installation distance output mode data start character
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
BBBB	4	X coordinate, displays the left and right positions based on the radar, unit cm First byte is sign, - is left, + is right,
CCCC	4	Z coordinate, represents the position before and after the radar. cm First byte is sign, - is backward, + is forward,
DDD	3	Amount of power in an object, Mark all values larger than 999 as 999
!	1	Character indicating the end of data.
$0!		When no object is detected, output at set time intervals
EXTENSION MODE		
$ABBBBCCCCDDDDEEE! (18 byte) (default mode)		
Item	**Byte**	**Content**
$	1	Wall installation distance output mode data start character
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
BBBB	4	X coordinate displays the left and right positions based on the radar, with the unit in cm. The first byte indicates the sign: '-' for left and '+' for right.
CCCC	4	The Y coordinate, based on the radar facing the ground, represents the distance information regarding the height from the radar to the object, with the unit in cm. The first byte indicates the sign, which is always '-'.
DDDD	4	The Z coordinate, based on the radar facing the ground, represents the distance information in the forward and backward directions from the radar, with the unit in cm. The first byte indicates the sign, with '-' for backward and '+' for forward.
EEE	3	Amount of power in an object, Mark all values larger than 999 as 999
!	1	Character indicating the end of data.
$0!		When no object is detected, output at set time intervals

■ Detection zone output mode



~ABC! (5 byte) (default mode)

Item	Byte	Content
~	1	Starting character indicating the detection zone output mode
A	1	Number distinguishing detected objects. e.g., 1: first object, 2: second object.
B	1	X-coordinate distance 0: Radar reference left: -400 to cm 1: Radar reference left: -300 to -399 cm 2: Radar reference left: -200 to -299 cm 3: Radar reference left: -100 to -199 cm 4: Radar reference left: -0 to -99 cm 5: Radar reference right: 1 to 99 cm 6: Right side based on radar: 100 to 199 cm 7: Radar reference right: 200-299 cm 8: Right side based on radar: 300-399 cm 9: Right side of radar: 400 to cm
C	1	Z-coordinate, the distance 0: Rear of radar reference: -400 to cm 1: Rear of radar standard: -300 to -399 cm 2: Rear of radar standard: -200 to -299 cm 3: Rear of radar standard: -100 to -199 cm 4: Rear of radar standard: -0 to -99 cm 5: Radar reference forward: 1 to 99 cm 6: Radar reference forward: 100 to 199 cm 7: Radar reference front: 200-299 cm 8: Radar reference front: 300-399 cm 9: Radar reference front: 400~cm
!	1	Character indicating the end of data.
~0!		When no object is detected, output at set time intervals

7.3 Wall mounting occupancy detection mode & Celling mounting occupancy detection mode (mode3 & mode4)

- Displays the approximate number of people and occupancy areas in spaces such as meeting rooms or hallways.
- The occupancy detection mode offers two options: a basic mode that outputs the approximate number of people within the detection area, and an area output mode that allows you to set zones within the detection area and outputs the approximate number of people in each zone.
- Each mode supports two type duration options: normal duration mode and short duration mode. The normal duration mode is recommended for environments where people stay for an extended period, such as meeting rooms. The short duration mode is recommended for environments where people pass through briefly, such as hallways. The default setting is the normal duration mode.

- The basic detection distance settings for wall installation are as follows:

x_min	-4,000 mm	x_max	4,000 mm
y_min	-2,000 mm	y_max	1,500 mm
z_min	400 mm	z_max	8,000 mm
alrmin	0 °	alrmax	180 °
adumin	20 °	adwmzx	160 °
Z always has + values			

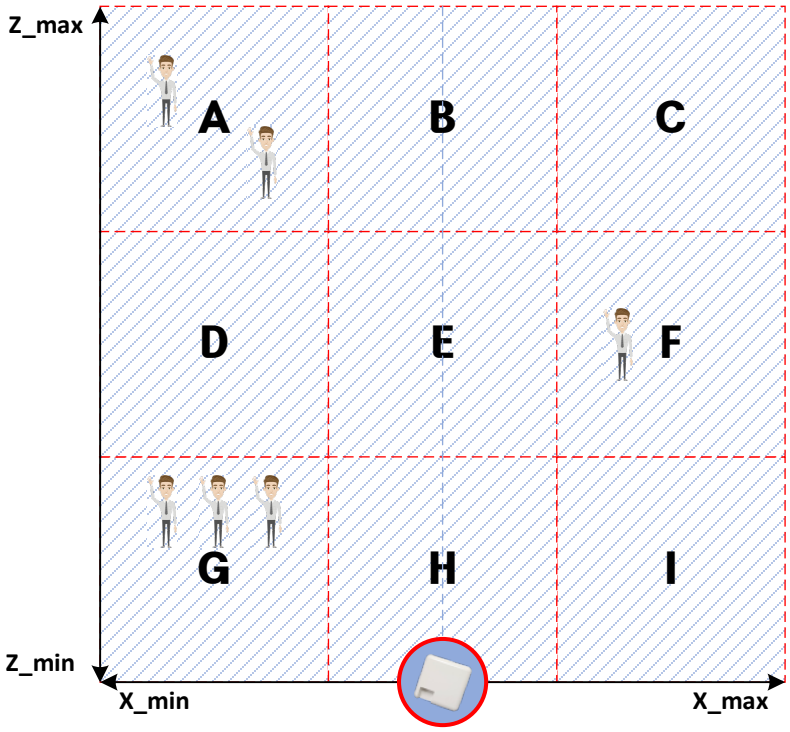
- The smaller the object, the detectable angle becomes narrower.
- The detection distance may be shortened depending on the installation environment, as well as the person's location or posture.

- The basic detection distance settings for ceiling installation are as follows:

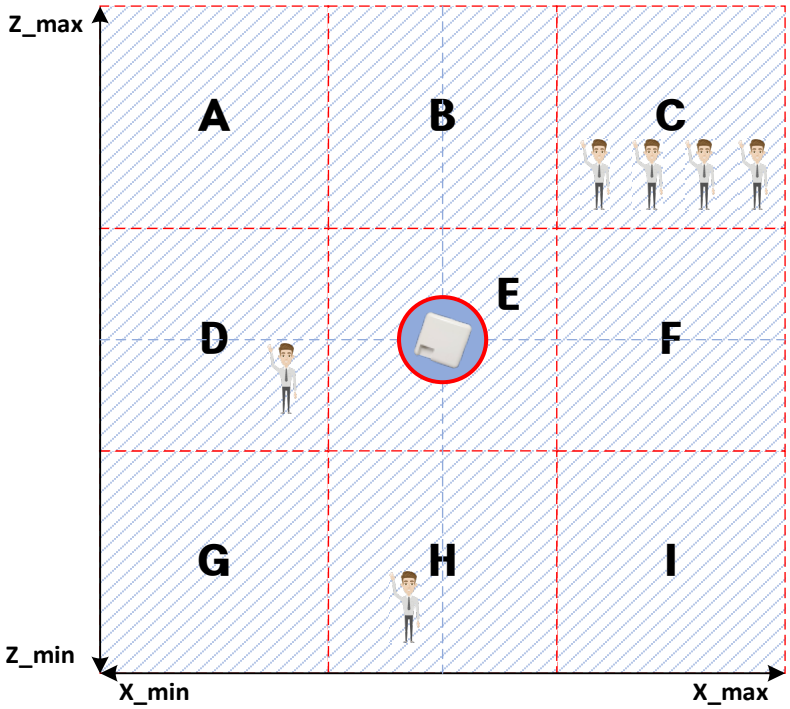
x_min	-3,000 mm	x_max	3,000 mm
y_min	-3,000 mm	y_max	-400 mm
z_min	-3,000 mm	z_max	3,000 mm
alrmin	15 °	alrmax	165 °
adumin	15 °	adwmzx	165 °
Y always has – values			

- The smaller the object, the detectable angle becomes narrower.

- Wall mounting occupancy detection area output mode



- Ceiling mounting occupancy detection area output mode



- The detection result may differ from the actual number of people by 1 to 3 individuals, providing only an approximate count.
- If they are close together, they may be recognized as one person.
- This function cannot determine the exact number or location of individuals and is not intended for use in counting people.

■ Occupancy detection normal mode

<A! (3 byte) (default mode)		
Item	**Byte**	**Content**
<	1	Starting character indicating the occupancy detection
A	1	Number of people detected in space, up to 15 objects, default set 10
!	1	Character indicating the end of data.
<0!		When no object is detected, output at set time intervals

- For use in controlling the lighting, if a person is detected in the absence of others, it immediately outputs <1! and then outputs the detected value in the next output.
- In occupancy detection mode, the data is output approximately once every 15 seconds.

■ Occupancy detection zone output mode

>ABCDEFGHI! (11 byte)		
Item	**Byte**	**Content**
>	1	Starting character indicating the occupancy detection
A	1	Number of people detected in Area A
B	1	Number of people detected in Area B
C	1	Number of people detected in Area C
D	1	Number of people detected in Area D
E	1	Number of people detected in Area E
F	1	Number of people detected in Area F
G	1	Number of people detected in Area G
H	1	Number of people detected in Area H
I	1	Number of people detected in Area I
!	1	Character indicating the end of data.
>0!		When no object is detected, output at set time intervals

7.4 Enable Static Object Detection in Distance-Based Measurement Mode

- Distance-based measurement mode (mode 1, mode 2, mode 3, mode 4) allows the default state to detect only objects with movement.
- To detect static objects in distance-based measurement mode, use the '**static**' command to enable fixed object detection.
- When you activate the static object detection function, when there is an object with motion, it first outputs the object with motion detected. It detects fixed objects only when there is no object with motion detected.
- The static object detection function does not know the direction information of the detected object and only outputs the distance to the object in z coordinates.

- **Static object detection should be used in a limited way.**
- **By activating the static object detection function, you can always output detection results depending on the surrounding environment and continue to output results that you do not want.**

7.5 Side wall mounting vital-signal measurement mode (mode 5)

- Dedicated mode for vital signal detection, with a detection range of up to 2.5m.
- The basic detection distance settings are as follows:

x_min	-1,250 mm	x_max	1,250 mm
y_min	-1,500 mm	y_max	1,000 mm
z_min	400 mm	z_max	2,500 mm
alrmin	30 °	alrmax	150 °
adumin	30 °	adwmzx	150 °
Z always has + values			

- The sensor detects and outputs the vital signs of the person closest to it.
- The data provided consists of respiratory rate, heart rate, HRV, respiratory waveform, heart rate waveform, and optional data. Optional data can be selected from detection distance or power, with the default set to detection distance.
- The respiration rate measured in a stable state has an accuracy of over 85%, while the heart rate is around 85%.
- This mode supports data extension mode. The data extension mode is activated with the command 'exten 1\n' and deactivated with the command 'exten 0\n'.
- When the extension mode is activated, it provides the following additional data

Normal mode	Vital status Heart rate Breath rate Heart wave data Breath wave data Option data: distance or power
Extension mode	It includes all the data from the normal mode, along with additional information. X-axis distance information Y-axis distance information Z-axis distance information Straight-line distance information Power data

- **Detection is not possible or accuracy is low in the following cases:**
  - The radar detects vital signs by sensing the small movements of the chest that occur during breathing or heartbeat. If you intentionally breathe rapidly or hold your breath, the chest becomes constrained, and vital signs may not be detected properly. In particular, under these conditions, the heartbeat may not be detected.
  - The principle by which the sensor detects vital signs is by detecting the frequency of small movements of the chest back and forth when a person breathes and the heart beats.
  - The best detection performance can be achieved when in a stable state. A stable state is when your hands are down, your chest is facing the direction of the radar sensor, and you are not moving or talking.
  - When arms are crossed, hands are raised in front of the chest while sitting at a desk (e.g., typing on a keyboard), or hands cover and move in front of the chest, detection may fail, or normal values may not be output.
  - If there is a moving object between the sensor and the person being sensed, or if the object is closer than the person, normal data cannot be output.
  - If the sensor is installed on the back or arm, it can detect vital signs, but the accuracy is lower.
  - If there is a moving object, such as a fan or robot vacuum cleaner, in the detection area, a detection signal may be output.
  - It is not recommended to use the sensor if there are complicated objects nearby.

- Data Output Format : ASCII
  - ■ No waveform mode : Default mode

NORMAL MODE		
%ABBBCCDDD! (11 byte) (default mode)		
Item	**Byte**	**Content**
%	1	Starting character indicating the output of vital sign data
A	1	Vital sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal 4: Fall detecttion
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DDD	3	Option data, distance(cm) or power, Default setting is distance
!	1	Character indicating the end of data.
%0!		When no object is detected, output at set time intervals
EXTENSION MODE		
%ABBBCCDDDDEEEEFFFGGG! (22 byte)		
Item	**Byte**	**Content**
%	1	Starting character indicating the output of vital-sign data
A	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal 4: Fall detecttion
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DDDD	4	X coordinate, displays the left and right positions based on the radar, unit in cm First byte is sign, - is left, + is right,
EEEE	4	Y coordinate, displays the below and above positions based on the radar, unit in cm First byte is sign, - is below, + is above,
FFF	3	Z coordinate, displays the front positions based on the radar, unit cm
GGG	3	Power
!	1	Character indicating the end of data.
%0!		When no object is detected, output at set time intervals

- The initial state of this mode is the basic data mode without vital sign waveform data. To include vital sign waveform data, the 'onwave\n' command must be used, and to output extended data, the 'exten 1\n' command must be used for additional settings.

■ Waveform mode : 10 outputs per second

NORMAL MODE		
@ABBBCCDEEEFFF! (15 byte)		
Item	**Byte**	**Content**
@	1	Starting character indicating the output of vital sign data
A	1	Vital sign detection status 0: No target 1: Target, Moving 2: Target, No vital -signal 3: Target, Vital-signal 4: Fall detecttion
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
D	1	Heart rate pulse, 0, 1
EEE	3	Respiration waveform data, 0 ~ 999
FFF	3	Option data, distance(cm) or power, Default setting is distance
!	1	Character indicating the end of data.
@0!		When no object is detected, output at set time intervals
EXTENSION MODE		
@ABBBCCDEEEFFFFGGGGHHHIII! (26 byte)		
Item	**Byte**	**Content**
@	1	Starting character indicating the output of vital-sign data
A	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal 4: Fall detecttion
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
D	1	Heart rate pulse, 0, 1
EEE	3	Respiration waveform data, 0 ~ 999
FFFF	4	X coordinate, displays the left and right positions based on the radar, unit cm First byte is sign, - is left, + is right,
GGGG	4	Y coordinate, displays the below and above positions based on the radar, unit in cm First byte is sign, - is below, + is above,
HHH	3	Z coordinate, displays the front positions based on the radar, unit in cm
III	3	Power
!	1	Character indicating the end of data.
@0!		When no object is detected, output at set time intervals

7.6 Wall mounting vital-signal and position measurement combined mode (mode 6)

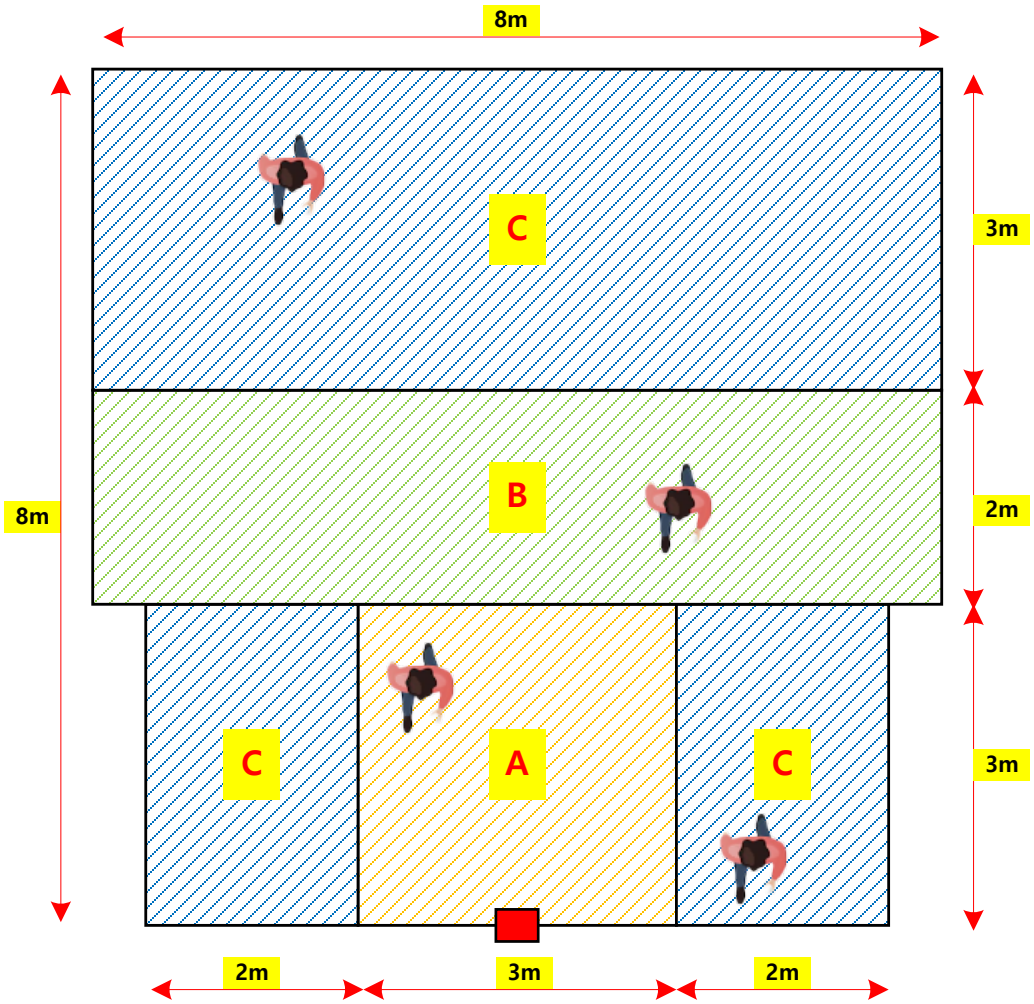
- A complex mode that outputs both a person's vital signs and location information, which can only be installed on a wall.

- The basic detection distance settings are as follows:

x_min	-4,000 mm	x_max	4,000 mm
y_min	-2,000 mm	y_max	1,500 mm
z_min	400 mm	z_max	8,000 mm
alrmin	0 °	alrmax	180 °
adumin	20 °	adwmzx	160 °
Z always has + values			

- The smaller the object, the narrower the detection angle.

- Vital signal information can only be collected from the person closest to the sensor. If there is a moving object closer to the sensor than the person whose vital signals are being detected, the vital signals will not be detected.
- Detection area:
- Among the detected biometric information, the breathing rate measured in a stable state within a space of 3 meters has an accuracy of over 80%. Heart rate information is approximately 80% accurate.



- Area A: Heart rate, breathing information, location information
- Area B: Respiration information, location information
- Area C: Location information

- **Detection is not possible or accuracy is low in the following cases:**
  - Depending on the installation environment and the type of object being detected, the detectable distance may be shorter than the set detection distance.
  - The principle by which the sensor detects vital signs is by detecting the frequency of small movements of the chest back and forth when a person breathes and the heart beats.
  - It performs best in areas without objects such as furniture, such as conference rooms or hallways. If there are complex objects in the detection area, detection may not work well or may result in incorrect readings. It cannot distinguish between people and moving objects. When a person is nearby, they may be recognized as a single individual, and even when a fan or air conditioner is operating, it may be recognized as an object.

- Data Output Format : ASCII
  - ■ Distance output mode

&ABBBBCCCDDDEFFFGG! (19 byte) (default mode)		
Item	**Byte**	**Content**
&	1	Vital sign and distance composite signal data start character
A	1	Number distinguishing detected objects, up to 15 objects, default set 3 e.g., 1: first object, 2: second object.
BBBB	4	X coordinate, displays the left and right positions based on the radar, unit cm First byte is sign, - is left, + is right Mark all values larger than 999 cm as 999
CCC	3	Z coordinate, indicates the position in front and behind the radar, unit cm Mark all values larger than 999 cm as 999
DDD	3	power
E	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal 4: Fall detecttion
FFF	3	heart rate per minute
GG	2	respiratory rate per minute
!	1	Character indicating the end of data.
&0!		When no object is detected, output at set time intervals

- It does not display the sign because it only outputs the positive distance in the Z direction.

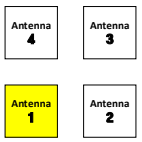
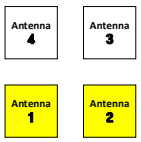
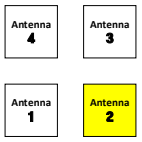
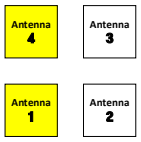
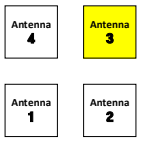
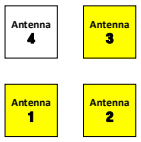
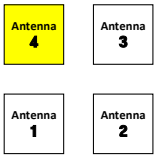
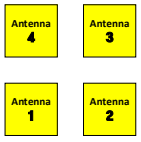
  - ■ Zone output mode

?ABBBBCCCDDDEFFFGGHIII! (23 byte)		
Item	**Byte**	
?	1	
A	1	Number distinguishing detected objects, up to 15 objects, default set 3 e.g., 1: first object, 2: second object.
BBBB	4	X coordinate, displays the left and right positions based on the radar, unit cm First byte is sign, - is left, + is right Mark all values larger than 999 cm as 999
CCC	3	Z coordinate, indicates the position in front and behind the radar, unit cm Mark all values larger than 999 cm as 999
DDD	3	power
E	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal 4: Fall detecttion
FFF	3	heart rate per minute
GG	2	respiratory rate per minute
H	1	Heart rate pulse, 0, 1
III	3	Respiration waveform data, 0 ~ 999
!	1	Character indicating the end of data.
?0!		When no object is detected, output at set time intervals

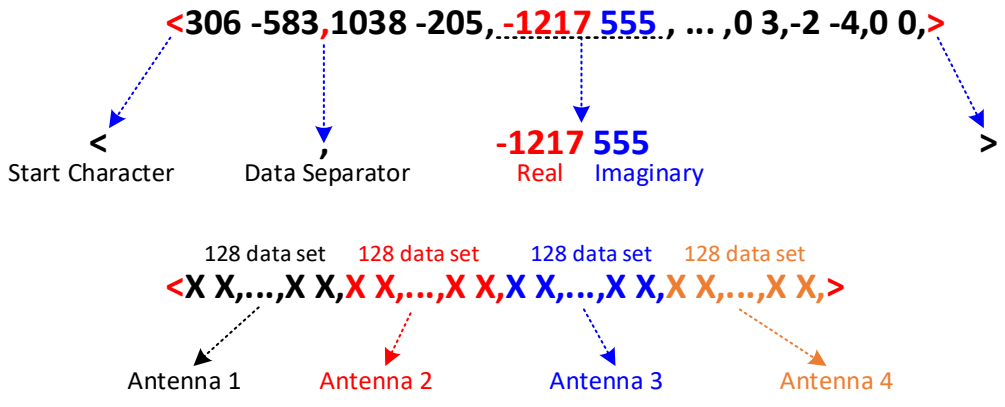
- The initial state of this mode is the basic data mode without vital sign waveform data. To include vital sign waveform data, the 'onave\n' command must be used.

7.7 Option mode: Raw Data Output Mode (mode 10)

- Can be activated and used if an additional license is available.
- A mode that directly outputs the Real and Imaginary data of each MMIC antenna.
- Depending on the output options, raw data from a single antenna or multiple antenna pairs can be output.
- Each antenna contains 128 pairs of Real and Imaginary data, and each data value is a signed integer.

Command	Data count	Output antenna	Command	Data count	Output antenna
raw 1	128		raw 5	256	
raw 2	128		raw 6	256	
raw 3	128		raw 7	384	
raw 4	128		Raw 8	512	

- Data Structure



- The number of output data points varies depending on the number of selected antennas.
- The output sequence follows the antenna order, continuously outputting 128 pairs of data for each selected antenna.

7.8 Option mode: Advanced Vital Sign mode (mode 11)

- Can be activated and used if an additional license is available.
- The Intelligent Vital Sign Detection Mode is an advanced mode based on the basic Vital Sign Detection Mode (Mode 5). It includes RMSSD and SDNN data, which are components of HRV.
- RMSSD, which evaluates parasympathetic nervous system activity (heart stability and recovery) and measures short-term heart rate variability, outputs results when vital signs are continuously detected for more than 30 seconds in a stable state.
- SDNN, which reflects the balance between the sympathetic and parasympathetic nervous systems, begins outputting data when vital signs are detected for more than 30 seconds in a stable state. SDNN is based on short-term monitoring and stabilizes to normal values after maintaining a stable state for at least 5 minutes.

- The basic detection distance settings are as follows:

x_min	-1,250 mm	x_max	1,250 mm
y_min	-1,500 mm	y_max	1,000 mm
z_min	400 mm	z_max	2,500 mm
alrmin	30 °	alrmax	150 °
adumin	30 °	adwmzx	150 °
Z always has + values			

- This mode supports data extension mode. The data extension mode is activated with the command 'exten 1\n' and deactivated with the command 'exten 0\n'.
- When the extension mode is activated, it provides the following additional data

Normal mode	Vital status Heart rate Breath rate RMSSD data SDNN data Heart wave data Breath wave data Option data : distance or power
Extension mode	It includes all the data from the normal mode, along with additional information. X-axis distance information Y-axis distance information Z-axis distance information Straight-line distance information Power data

- Data Output Format : ASCII
  - ■ Waveform mode : Default mode : 10 outputs per second

NORMAL MODE		
{ABBBCCDDEEFGGHHH! (18 byte)		
Item	**Byte**	**Content**
{	1	Starting character indicating the output of vital sign data
A	1	Vital sign detection status 0: No target 1: Target, Moving 2: Target, No vital -signal 3: Target, Vital-signal
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
F	1	Heart rate pulse, 0, 1
GG	2	Respiration waveform data, 0 ~ 99
HHH	3	Option data, distance(cm) or power, Default setting is distance
!	1	Character indicating the end of data.
{0!		When no object is detected, output at set time intervals
EXTENSION MODE		

{ABBBCCDDEEFGGHHHHIIIIJJJKKK! (29 byte)		
Item	**Byte**	**Content**
{	1	Starting character indicating the output of vital-sign data
A	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
F	1	Heart rate pulse, 0, 1
GG	2	Respiration waveform data, 0 ~ 99
HHHH	4	X coordinate, displays the left and right positions based on the radar, unit cm First byte is sign, - is left, + is right,
IIII	4	Y coordinate, displays the below and above positions based on the radar, unit in cm First byte is sign, - is below, + is above,
JJJ	3	Z coordinate, displays the front positions based on the radar, unit in cm
KKK	3	Power
!	1	Character indicating the end of data.
{0!		When no object is detected, output at set time intervals

■ No waveform mode

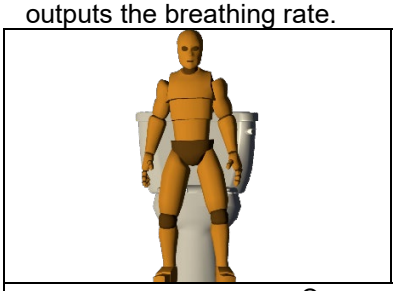
NORMAL MODE		
}ABBBCCDDEEFFF! (15 byte) (default mode)		
Item	**Byte**	**Content**
}	1	Starting character indicating the output of vital sign data
A	1	Vital sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFF	3	Option data, distance(cm) or power, Default setting is distance
!	1	Character indicating the end of data.
}0!		When no object is detected, output at set time intervals
EXTENSION MODE		
}ABBBCCDDEEFFFFGGGGHHHIII! (26 byte)		
Item	**Byte**	**Content**
}	1	Starting character indicating the output of vital-sign data
A	1	Vital-sign detection status 0: No target 1: Target, Moving 2: Target, No vital-signal 3: Target, Vital-signal
BBB	3	heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFFF	4	X coordinate, displays the left and right positions based on the radar, unit in cm First byte is sign, - is left, + is right,
GGGG	4	Y coordinate, displays the below and above positions based on the radar, unit in cm First byte is sign, - is below, + is above,
HHH	3	Z coordinate, displays the front positions based on the radar, unit cm
III	3	Power
!	1	Character indicating the end of data.
}0!		When no object is detected, output at set time intervals

- The initial state of this mode is the basic data mode without vital sign waveform data. To include vital sign waveform data, the 'onwave\n' command must be used, and to output extended data, the 'exten 1\n' command must be used for additional settings.

7.9 Option mode: Restroom ceiling mounting mode (mode 12)

- Can be activated and used if an additional license is available.

- It is installed on the bathroom ceiling and detects the presence of a person inside the bathroom. When a person enters, it determines occupancy, and if a fallen state is detected, it outputs that status. Additionally, when the person is sitting on the toilet or lying down, it detects breathing and outputs the breathing rate.

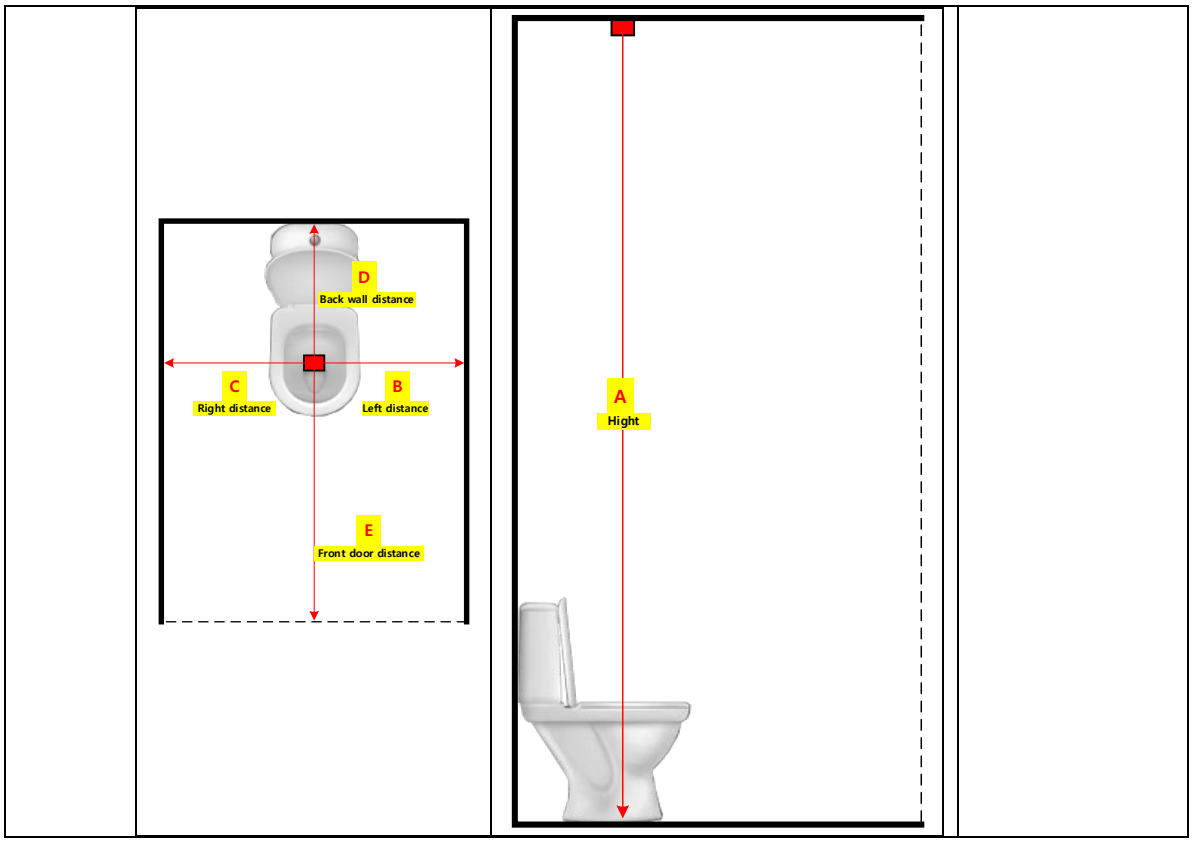
<table>
<tr><td colspan="2">  </td><td>  </td></tr>
<tr><td colspan="2">Occupancy state</td><td>Lying down state</td></tr>
</table>

- The sensor is installed on the ceiling, facing the person's head. Therefore, the accuracy of the detected heart rate and respiration information is not guaranteed and should be used for reference purposes only.

- The basic detection distance settings are as follows:

x_min	-500 mm	x_max	500 mm
y_min	-2,200 mm	y_max	-200 mm
z_min	-400 mm	z_max	1,000 mm
alrmin	50 °	alrmax	130 °
adumin	50 °	adwmzx	130 °
Y always has - values			

- For installation in a larger space, please contact **Inzinious, Inc.,** headquarters.
- To install the sensor in the bathroom and ensure its proper operation, the following setting values must be entered:



- A: Sensor installation height (setting command: y_min, always -)
- B: Distance from the sensor to the left wall (setting command: x_min)
- C: Distance from the sensor to the right wall (setting command: x_max)
- D: Distance from the sensor to the back wall (setting command: z_min)
- E: Distance from the sensor to the front door (setting command: z_max)
  Be sure to install the Z_MAX orientation towards the door.

- **Detection is not possible or accuracy is low in the following cases:**
  - If the distance settings are not accurate, the system will not operate properly.
  - If more than one person is present, the system will not function correctly.

- Data Output Format: ASCII

^A! **(If A is 0, 1, 4)**
^ABB! (5 byte) (default mode)

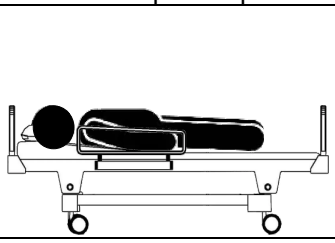
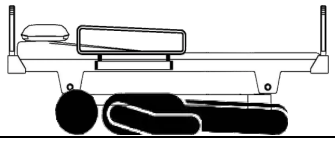
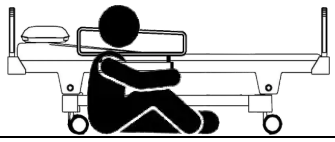
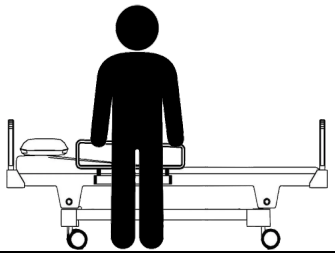
Item	Byte	Content
^	1	The starting character indicating the occupancy status output in restroom mode.
A	1	Occupancy detection status 0 : Rest room Empty 1 : - 2 : Present in the rest room 3 : lying down in the rest room 4 : Warning : Occurs when a person approaches near the ceiling
BB	2	Respiration rate per minute
!	1	Character indicating the end of data.
^0!		When no object is detected, output at set time intervals

7.9 Option mode: Celling mounting hospital bed mode (mode 13)

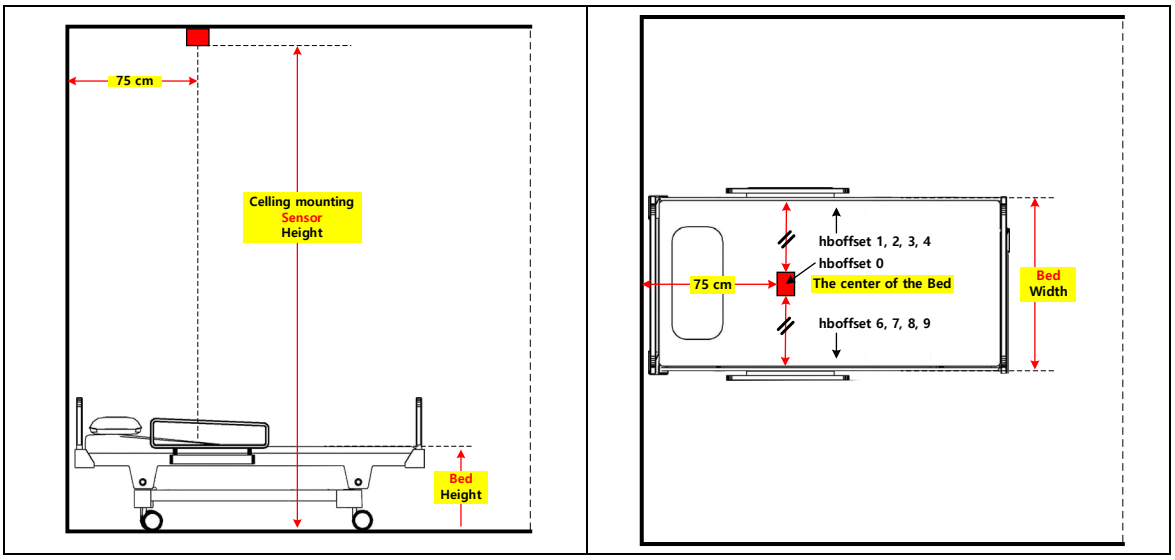
- Can be activated and used if an additional license is available.

- Installed on the ceiling above the chest position of the hospital bed, it performs functions such as detecting the presence of patients, their posture, vital signs (heart rate, respiratory rate, HRV), fall risk detection and warning, and pressure ulcer risk detection and warning.

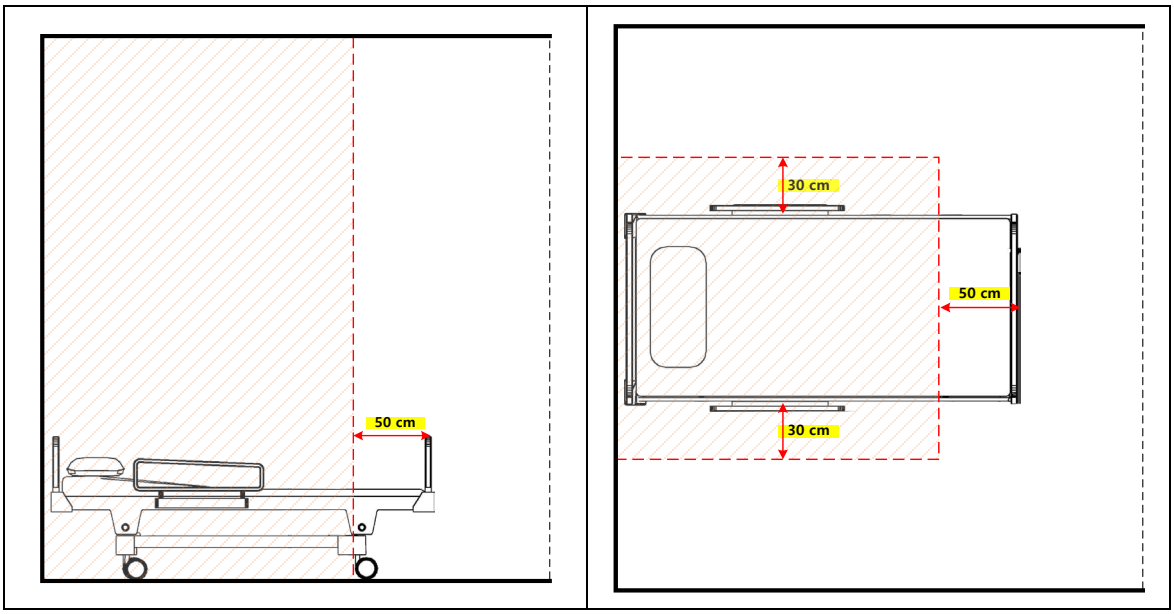
- The detectable patient postures are as follows:

<table>
<tr><td colspan="2">  </td><td></td></tr>
<tr><td colspan="2">Person present on the bed</td><td></td></tr>
<tr><td>  </td><td>  </td><td>  </td></tr>
<tr><td colspan="2">Sitting or lying next to the bed (outside the bed area)</td><td>Standing position next to the bed outside</td></tr>
</table>

- The installation location of the AS_614 is as follows:



  - In the ceiling installation mode, when the bed is against the wall with the head facing the wall, the sensor should be installed 75 cm away from the wall.
  - The sensor must be positioned exactly at the center of the bed along the left-right direction. If the bed is moved within a 20 cm range to the left or right after sensor installation, the offset should be adjusted using the 'hboffset' command.
- Area for detecting patients on and around the bed



- The detection area includes all areas inside the bed and extends 30 cm to the sides outside the bed to prevent interference from other patients or medical personnel moving outside the bed.

- The fall risk warning is activated when the patient is detected to be lying or sitting in the center of the bed and then sitting on the left or right side, indicating an intention to get up from the bed and move outside.
- The fall occurrence alarm is activated when the patient is detected to be sitting or lying on the side floor outside the bed while in a fall risk state.
- The fall is detected only when the patient is lying on the bed and then sits at the edge of the bed before sitting or lying on the floor beside the bed. Falls are not detected in any other situations.

- **If the patient is lying in bed, the waveform is output according to breathing, heart rate, and options. The patient can take various postures, such as placing their hands on their chest or lying on their side. The radar does not determine the posture of a patient lying in bed, so it is unknown whether the detected heart rate was measured in a normal state. Therefore, the heart rate can only be measured and used for reference. On the other hand, breathing or waveforms provide more accurate values than heart rate. In the current version, the waveform of the biological signal only provides breathing.**

- The basic detection distance settings are as follows:

x_min	-750 mm	x_max	750 mm
y_min	-2,200 mm	y_max	0 mm
z_min	-750 mm	z_max	1,400 mm
alrmin	45 °	alrmax	135 °
adumin	45 °	adumax	150 °
Sensor height	2,200 mm	Bed height	550 mm
		Bed width	900 mm
Y always has - values			

- The bed width setting range is from a minimum of 900 mm to a maximum of 1,600 mm.

- Data Output Format: ASCII
  - ■ No waveform mode: default mode

/A! (If A is 0, 1, 2, 7, 8, 9)
/ABBBCCDDEEFFF! (15 byte) (default mode)

Item	Byte	Content
/	1	Starting character indicating the output of without vital-sign wave data
A	1	Hospital bed state 0: Bed empty 1: Patient standing on the side outside the bed 2: Patient sitting or lying on the side outside the bed 3: Patient on the bed (awake state) 4: Patient on the bed (tossing and turning) 5: Patient on the bed (sleeping state) 6: **Patient on the bed (Bed fall risk warning)** 7: Bed fall occurrence detection 8: Pressure ulcer risk warning 9: High pressure ulcer risk warning
BBB	3	Heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFF	3	Temporary data
!	1	Character indicating the end of data.
/0!		When no object is detected, output at set time intervals

  - ■ Include waveform mode: 10 outputs per second

\A! (If A is 0, 1, 2, 7, 8, 9)
\ABBBCCDDEEFFFGGG! (18 byte) (default mode)

Item	Byte	Content
\	1	Starting character indicating the output of vital-sign wave data
A	1	Hospital bed state 0: Bed empty 1: Patient standing on the side outside the bed 2: Patient sitting or lying on the side outside the bed 3: Patient on the bed (awake state) 4: Patient on the bed (tossing and turning) 5: Patient on the bed (sleeping state) 6: **Patient on the bed (Bed fall risk warning)** 7: Bed fall occurrence detection 8: Pressure ulcer risk warning 9: High pressure ulcer risk warning
BBB	3	Heart rate per minute
CC	2	respiratory rate per minute
DD	2	HRV: RMSSD (ms), 0 ~ 99
EE	2	HRV: SDNN (ms), 0 ~ 99
FFF	3	Respiration waveform data, 0 ~ 999
GGG	3	Temporary data
!	1	Character indicating the end of data.
\0!		When no object is detected, output at set time intervals

7.10 Output data start and end character for each mode

Order	Mode	Data output start character
1	**Mode 1 :** Wall mounting position measurement mode	# : Distance output mode * : Zone output mode
2	**Mode 2 :** Ceiling mounting position measurement mode	$: Distance output mode ~ : Zone output mode
3	**Mode 3:** Wall mounting occupancy detection mode	< : General Output Mode > : Area Output Mode
4	**Mode 4:** Celling mounting occupancy detection mode	< : General Output Mode > : Area Output Mode
5	**Mode 5:** Wall or side mounting vital-signal measurement mode	@ : Including vital sign waveform output mode % : Without vital sign waveform mode
6	**Mode 6:** Wall mounting vital-signal and position measurement combined mode	& : Distance output mode ? : Zone output mode
7	**Mode 10:** Raw data output mode	[: Start character] : End character
8	**Option Mode 11:** Advanced vital sign mode	{ : Including vital sign waveform output mode } : Without vital sign waveform mode
9	**Option Mode 12:** Restroom ceiling mounting mode	^
10	**Option Mode 13:** Celling mounting hospital bed mode	\ : Including vital-signal waveform output mode / : Without vital-sign waveform mode
11	Response to Radar Description Command	=
12	End character of all packets	!

8 Protocol

- AS_614 supports 3.3V level UART communication, and 1.8V, 3.3V, and 5V signal interfaces are possible.
- It utilizes ASCII character codes for sending and receiving data and supports Command Line Input (CLI).
- The LED turns on when an object is detected.
- UART Settings:

Item	Setting value
Baud rate	230,400 bps
Data	8 bit
Parity bit	None
Stop bits	1 bit
Flow control	None

- All commands must be in lowercase.

- Changed commands retain their values even after the power is turned on again.
- All commands must include LF (line feed/newline) or CR (carriage return) when transmitted.
- The starting character of the response to a command is '=' and the ending character is '!'.
  For example, the response to the wait command is =15!
- **Be sure to proceed with the reset command when changing the mode or making a new setting.**
  If you continue to change the setting without initializing, detection may proceed to a different state, and the result may be output differently from the actual value due to the previously set value.
- When sending commands consecutively, they must be sent at intervals of more than 500 ms, during which you can reset the system but should not set any other commands.

- When sending multiple commands sequentially, a time interval of at least 500ms is required between each command.

9 Command

Command	Example	Default value / Setting range	Explanation
Flow Control Commands			
start	start	-	Data Output Start Default Start State When restarting with the start command after a stop command, the buffer maintains its previous state and starts from there, so normal values may not be output initially.
stop	stop	-	**Data Output Stop** Only data output is stopped, while internal object detection continues to operate.
reset	reset	-	Restarts the module via software while maintaining the settings.
hreset	hreset	-	Restarts the system while maintaining settings, equivalent to reconnecting power.
freset	freset	-	**Factory Reset** Factory reset, restores all settings to initial state. When changing various setting values for testing purposes, unintended behavior often occurs due to previously configured settings being retained while setting new modes or configuration states. Therefore, it is recommended to initialize the system with the freset command before applying new settings.
Mode and Configuration Control Commands			
mode	mode mode 2	1 1 ~ 6, 11, 12	**Sensor Operation Mode Setting** 1: Wall-mounted position detection mode 2: Ceiling-mounted position detection mode 3: Wall-mounted occupancy detection mode 4: Ceiling-mounted occupancy detection mode 5: Wall or side-mounted vital signs measurement mode 6: Wall-mounted combined vital signs and position measurement mode 10: Raw data output mode (optional mode) 11: Advanced vital signs mode (optional mode) 12: Bathroom ceiling-mounted mode (optional mode) 13: Hospital bed ceiling-mounted mode (optional mode) Optional modes are activated only when a license is available. When the mode is changed, x_min, x_max, y_min, y_max, z_min, z_max values are reset to default. The first output data after mode change may not show normal values as it is based on data from the previous mode.
exten	exten exten 1	0 0, 1	0: Basic data output 1: Extended data output: Additional data varies depending on the mode
holdtime	holdtime holdtime 2	1 1 ~ 10	Sets the duration for maintaining status even after an object temporarily disappears. Used to prevent situations where objects with small movements are temporarily undetected. When the value is large, the object status is maintained for an extended period even after the object has disappeared.
led	led led 0	0 0, 1	Sets the LED activation state 0: Enable LED output 1: Disable LED
opt	opt opt 1	1 0, 1, 2, 3	Sets selectable data types 0: No data output 1: Object distance, default value 2: Object reflection magnitude 3: Object movement magnitude
outtime	outtime outtime 1	0 0, 1, 2, 3	Data output time interval setting: 0: Initialize to default settings for each mode 1: Output all data detected in real-time 2: Output once per second 3: Output only when there are changes in output data (occupancy mode only) When no object is detected, the time set in the standby time is output. In occupancy mode, the time set in the standby time is always output.
range	range range 50	40 20 ~ 150	Sets the distance for judging objects detected at close range as a single object If the value is too small, false detections may occur; if the value is too large, multiple objects close to each other will be output as a single object
response	response response 1	1 0, 1, 2, 3, 4	Sets the initial detection response time 0: Fast detection, may respond to noise 1: Default value 4: Favorable against noise, does not respond to small or fast movements
setnum	setnum setnum 10	3 1 ~ 5	Sets the number of detected objects Cannot be set to a fixed value in modes that include vital signs.
static	static static 2	0 0, 1, 2, 3, 4, 5	Enable the ability to detect static objects in distance-based mode 0: Disable fixed object detection (default state) 1: Set fixed object detection area to 1m 2: Set fixed object detection area to 2m 3: Set fixed object detection area to 3m 4: Set fixed object detection area to 4m 5: Set fixed object detection area to 5m
pthold	pthold pthold 10	4 3 ~ 500	Sets the detection level based on the reflected power of objects. Lower values increase sensitivity to small movements but also increase noise. Increase the value if sensitive to noise or to detect only large objects. The default value varies by mode, and the setting range is from 4 to 500. **CAUTION:** If the value is set too high, object detection may fail..
trace	trace trace 0	0 0, 1	Enables tracking function when outputting detected objects 0: Output object numbers starting from the closest object, default state 1: Maintain detected object numbers
user	user user 27	- 0 ~ 65535	User-defined data, allows users to store arbitrary data. The value is retained after reset until changed by the user.

			For example, product serial number, etc.
ver	ver	-	Checks the firmware version ex) L25Aug16102.0! Firmware version 2.0 compiled on 2025.08.16 at 10:00
vitalhl	vitalhl	0 0, 1, 2	Set the level of continuous detection of fine vital signs in the mode of detecting vital signs Change settings in situations where there is no movement, such as sleep, and the bio-signal is weak 0: Default state 1: Small movements that are still in the chair 2: There is no movement, such as sleeping, and it is applied in a fine bio-signal state
wait	wait wait 360	15 3 ~ 43200	Time interval for outputting detection status when no object is detected Default interval is output once every 15 seconds Can be set from a minimum of 3 seconds to a maximum of 12 hours (43,200 seconds).
wave	wave wave 0	0 0, 1	Outputs respiratory waveform from vital signs 0: Output without respiratory waveform data, default output once per second 1: Output respiratory waveform data, output 10 times per second
zone	zone zone 0	0 0, 1	Outputs position data of detected objects as zone information 0: Output as distance values, default state 1: Output as zone value
Detection Area Configuration Commands			
xmin	xmin xmin 1000 (unit: mm)	- -5000 ~ 5000 (wal) -4000 ~ 4000 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions
xmax	xmax xmax 1000 (unit: mm)	- -5000 ~ 5000 (wall) -4000 ~ 4000 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions
ymin	ymin ymin 1000 (unit: mm)	- -5000 ~ 5000 (wall) -4000 ~ 0 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions. In ceiling-mounted mode, always has a negative (-) value.
ymax	ymax ymax 1000 (unit: mm)	- -5000 ~ 5000 (wall) -4000 ~ 0 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions n ceiling-mounted state, the setting is ignored and processed as 0 In ceiling-mounted mode, always set to a negative (-) value.
zmin	zmin zmin 1000 (unit: mm)	- 0 ~ 10000 (wall) -4000 ~ 4000 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions In wall-mounted mode, always set to a positive (+) value
zmax	zmax zmax 1000 (unit: mm)	- 0 ~ 10000 (wall) -4000 ~ 4000 (celling)	Coordinates according to installation position refer to Section 6. Radar Coordinate Directions In wall-mounted mode, always set to a positive (+) value
alrmin	alrmin alrmin 30 (uint: °)	- 0 ~ 180	Sets the minimum detection start angle where the left direction from the radar is set to 0° and the right direction is set to 180°
alrmax	alrmax alrmax 150 (uint: °)	- 0 ~ 180	Sets the maximum detection end angle where the left direction from the radar is set to 0° and the right direction is set to 180°
adumin	adumin adumin 30 (uint: °)	- 0 ~ 180	Sets the minimum detection start angle where the downward direction from the radar is set to 0° and the upward direction is set to 180°
adumax	adumin adumin 150 (uint: °)	- 0 ~ 180	Sets the maximum detection end angle where the downward direction from the radar is set to 0° and the upward direction is set to 180°
Debug-Related Commands			
help	help	-	Outputs the types of available commands
info	info	-	Outputs the configured detection area status
Optional Mode Exclusive Commands			
hbh	hbh hbh 500	550 300 ~ 700	Applicable only to mode 13 Distance from the floor to the top of the hospital mattress, unit is mm.
hbw	hbw hbw 900	900 900 ~ 1600	Applicable only to mode 13 Width of the mattress, unit is mm.
hboffset	hboffset hboffset 5	5 1, 2, 3, 4, 5, 6, 7, 8, 9	Applicable only to mode 13 After installing the sensor at the center of the bed in the left-right direction, sets the offset value to compensate for position when the bed moves within a range of 20cm to the left or right. When the sensor is positioned at the center of the bed, the default is 5. 1: When the reference bed with a person lying down has moved 20cm to the left 2: When the reference bed with a person lying down has moved 15cm to the left 3: When the reference bed with a person lying down has moved 10cm to the left 4: When the reference bed with a person lying down has moved 5cm to the left 5: When the sensor is installed at the center of the bed, default value 6: When the reference bed with a person lying down has moved 5cm to the right 7: When the reference bed with a person lying down has moved 10cm to the right 8: When the reference bed with a person lying down has moved 15cm to the right 9: When the reference bed with a person lying down has moved 20cm to the right
wallside	wallside wallsiee 0	0 0, 1, 2	Applicable only to mode 13 Sets the direction where the bed is positioned against the wall. 0: When the head of the bed faces the wall 1: When the left side of the bed faces the wall 2: When the right side of the bed faces the wall
oht	oht oht	0 0, 1	Applicable only to occupancy modes 3 and 4 Environment setting based on the speed of human movement

			0: Normal response, used when movement is not fast such as in conference rooms, default state 1: Fast response, used when people pass by quickly such as in corridors
raw	raw raw 1	1 1, 2, 3, 4, 5, 6, 7, 8	Applicable only to mode 10 1: Raw data output from antenna 1 2: Raw data output from antenna 2 3: Raw data output from antenna 3 4: Raw data output from antenna 4 5: Sequential output of raw data from antennas 1 and 2 6: Sequential output of raw data from antennas 1 and 4 7: Sequential output of raw data from antennas 1, 2, and 3 8: Sequential output of raw data from antennas 1, 2, 3, and 4

10 Order Information

AS_614_P/C/U		
AS_611	P	Installing 2.54mm pin header
	C	Installing connector
	U	Installing USB-C connector
ex) AS_614_U		Model with USB-C connector only installed
ex) AS_614_PU		Model with 2.54mm pin header and USB-C connector installed
ex) AS_614_CU		Model with connector and USB-C connector installed

Warning

The device operates wirelessly and can generate radio waves that can prevent the pacemaker or implantable medical device from functioning properly. Do not use the device within 20 cm of a pacemaker or other implantable medical device. When using a pacemaker or other implantable medical device, do not use the device without first consulting your doctor or the manufacturer of the pacemaker or other implantable medical device.

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